

RORBACKSNAS, Sweden

WMO: 023060

Lat: 61.133N

Lon: 12.817E

Elev: 465

StdP: 95.86

Time Zone: 1.00 (EUC)

Period: 82-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-24.7	-29.9	0.2	-27.7	-26.8	0.3	-24.6	10.4	-1.5	10.0	-1.0	0.2	0	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	25.3	15.8	23.2	14.7	21.0	13.4	16.3	23.8	15.2	21.9	14.2	19.8	2.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	10.1	17.0	12.4	9.5	16.2	11.5	8.9	15.8	47.3	23.9	44.1	22.1	41.2	20.0	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	7.2	6.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.8	-9.1	-7.5	-3.9	0.7	7.2	10.9	13.5	11.4	7.2	2.3	-3.8	-7.9
	DBStd	9.22	7.74	6.23	4.80	3.90	3.62	3.42	3.38	3.29	3.17	3.87	5.28	6.84
	HDD10.0	3253	591	490	432	281	103	28	7	20	94	238	414	556
	HDD18.3	6039	849	723	691	530	344	223	154	216	334	496	664	814
	CDD10.0	261	0	0	0	0	17	56	114	64	9	0	0	0
	CDD18.3	6	0	0	0	0	0	1	4	2	0	0	0	0
	CDH23.3	156	0	0	0	0	1	23	88	44	0	0	0	0
	CDH26.7	16	0	0	0	0	0	1	6	9	0	0	0	0
Wind	WSAvg	1.9	2.0	1.8	2.0	2.0	2.0	2.1	1.8	1.6	1.9	1.9	1.6	1.6
Precipitation	PrecAvg	780	52	37	33	44	64	85	89	106	76	76	66	53
	PrecMax	1089	118	63	72	105	169	156	169	164	131	161	186	102
	PrecMin	618	11	7	4	9	10	8	20	42	25	12	23	5
	PrecStd	94	26	15	16	28	35	35	31	35	29	35	37	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	5.7	7.3	16.2	23.3	26.6	27.7	27.6	17.2	10.9	8.3	3.8
		MCWB	2.3	3.5	3.2	9.2	13.4	15.8	16.9	17.0	11.9	8.6	7.3	2.6
	2%	DB	2.6	4.1	4.4	11.7	20.9	23.6	25.9	23.6	15.4	9.9	5.8	2.8
		MCWB	1.2	2.2	1.2	6.2	11.9	14.9	16.3	15.1	11.2	7.8	4.6	1.9
	5%	DB	1.3	2.0	3.0	9.2	18.0	21.7	24.1	20.3	13.7	8.9	4.5	1.6
		MCWB	0.0	0.8	0.9	4.5	10.6	13.6	15.2	13.8	10.5	7.0	3.5	1.0
	10%	DB	0.2	0.7	1.9	7.2	15.6	19.3	21.6	17.6	12.3	7.8	2.9	0.5
		MCWB	-0.7	-0.2	0.3	3.2	9.3	12.2	14.0	12.2	9.3	6.3	2.1	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	4.2	3.6	9.4	14.2	16.6	17.9	17.1	14.0	10.0	7.5	3.1
		MCDB	3.5	5.3	5.9	15.8	21.4	25.2	24.7	25.7	15.9	10.6	8.3	3.5
	2%	WB	1.5	2.5	2.2	6.7	12.9	15.2	16.7	15.6	12.6	8.4	4.8	2.1
		MCDB	2.3	3.6	3.3	11.2	18.7	22.3	24.3	21.9	14.2	9.2	5.5	2.5
	5%	WB	0.3	1.1	1.3	5.0	11.4	14.3	15.8	14.4	11.0	7.4	3.6	1.0
		MCDB	1.1	1.8	2.5	8.2	17.1	20.8	22.8	19.2	12.9	8.3	4.4	1.6
	10%	WB	-0.8	0.0	0.4	3.8	9.9	12.9	14.6	13.2	9.7	6.3	2.2	-0.1
		MCDB	0.0	0.6	1.6	6.1	15.0	18.3	20.9	16.6	11.6	7.5	2.9	0.4
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.9	8.4	8.4	10.7	10.6	10.8	9.2	8.4	6.2	6.1	7.0
		MCDBR	5.8	5.8	8.0	12.3	16.4	16.4	15.6	15.0	10.6	7.0	5.3	6.4
		MCWBR	5.4	5.0	5.9	7.5	8.3	8.2	7.6	7.2	6.5	5.2	4.8	6.2
	5% WB	MCDBR	5.8	5.2	6.4	10.3	14.3	14.1	13.7	13.6	7.4	4.6	5.0	5.9
		MCWBR	5.7	4.9	4.9	6.5	7.4	7.3	7.0	6.6	5.3	4.0	4.7	6.0
Clear-Sky Solar Irradiance	taub	0.221	0.268	0.296	0.334	0.340	0.333	0.352	0.342	0.312	0.296	0.241	N/A	
	taud	2.156	2.274	2.289	2.351	2.452	2.495	2.453	2.472	2.545	2.495	2.247	N/A	
	Ebn at Noon	540	729	828	862	885	896	867	847	818	701	508	N/A	
	Edh at Noon	36	62	87	100	98	96	98	89	69	51	33	N/A	
All-Sky Solar Radiation	RadAvg	0.28	0.94	2.48	4.01	4.70	5.26	4.74	3.70	2.42	1.09	0.34	0.16	
	RadStd	0.04	0.18	0.26	0.43	0.46	0.40	0.64	0.35	0.30	0.20	0.09	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page